

Class: XI 'Science'

DAY SHIFT 'SET-B'

P.M: 1

Time: 1:30 hrs.

Subject: Chemistry [3011]

Candidates are required to give their answers in their own words as far as practicable. The figure in the margin indicate full marks.

Attempt all the questions.

GROUP - 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS

- A reducing agent is a substance which can
 - accept electron & is oxidized
 - donate electron & is oxidized
 - accept electron & is reduced
 - donate electron & is reduced
- Which of the following acts as reducing agent only?
 - SO₂
 - H₂SO₄
 - H₂S
 - SO₃
- Oxidation number of Pb in PbO₂ is
 - +1, +2
 - +2, +3
 - +3, +4
 - +2, +4
- 16 gm of O₂ at NTP would occupy a volume of
 - 11.2 lit
 - 22.4 lit
 - 5.6 lit
 - 16 lit
- The density of NH₃ at 30⁰ C and 5 atm pressure is
 - 2.30 gm/lit
 - 3.42 gm/lit
 - 23 gm/lit
 - 30 gm/lit
- The curve plotted between P and V at constant temperature is called
 - Isotherm
 - Isobar
 - Isochore
 - Isomer
- The size of following species increase in the order
 - Mg⁺² < Na⁺ < F⁻ < Al
 - F⁻ < Al < Na⁺ < Mg⁺²
 - Al < Mg⁺² < F⁻ < Na⁺
 - Na⁺ < Al < F⁻ < Mg⁺²
- The outmost electronic configuration of the least electronegative element is
 - ns²np⁵
 - ns²np⁴
 - ns²np³
 - ns¹
- The correct order of ionization energy of C, N, O, F is
 - F > O > N > C
 - O > F > N > C

SHORT ANSWER QUESTIONS.

- Define the terms:
 - Effective Nuclear Charge
 - Periodicity in Periodic Table
 - Define ionisation potential. Describe any two factors that affect ionisation potential.
- Define isoelectronic species.
 - Compare size of (i) Na with Na^+ (ii) Cl with Cl^-
 - Why is the electron affinity of chlorine higher than that of fluorine?
- Derive combined gas equation.
 - Find the value of 'R' in SI unit.
 - What is the significance of 'R'?
- Write any two applications of Boyle's law.
 - A spherical balloon of 21 cm diameter is to be filled with hydrogen gas at NTP from a cylinder containing gas at 20 atm and $27^\circ C$. If the cylinder can hold 2.82 litre of water vapour at NTP, calculate the number of balloons that can be filled up.

GROUP - 'C' [1X8=8]LONG ANSWER QUESTION.

- Differentiate between oxidation number and valency.
 - Define auto-redox reaction with example.
 - Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent.
- $$KMnO_4 + KCl + H_2SO_4 \longrightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$$
- Find the oxidation number of
 - S in H_2SO_4
 - Cr in CrO_5